KINETIC BREACH:

A Gamified Training System for Log and System Investigation Skills

by

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DECLARATION

I hereby declare that this project report entitled

"KINETIC BREACH: A Gamified Training System for Log and System Investigation Skills"

is the result of my own work except for quotations and summaries which have been duly acknowledged.

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ABSTRACT

Cybersecurity education increasingly requires practical and engaging learning environments that enable students to develop investigative and analytical skills beyond traditional theoretical instruction. However, many existing learning approaches provide limited opportunities for learners to perform realistic system investigations and log analysis activities within safe and accessible environments. In addition, real cybersecurity laboratories often require complex infrastructure, significant technical resources, and extensive maintenance, making them less suitable for beginner learners.

This project presents KINETIC BREACH, a web-based gamified training platform designed to improve log analysis and system investigation skills through interactive cybersecurity scenarios. The system provides a simulated Linux terminal environment where learners investigate realistic incidents by analyzing logs, examining system artefacts, and identifying digital evidence using common command-line tools. To support learner engagement and practical skill development, the platform incorporates gamification features including scoring mechanisms, progression tracking, achievements, and performance evaluation.

A key feature of the system is the integration of an adaptive Artificial Intelligence (AI) tutor known as ARIA. The AI component analyses learner behaviour, command history, and investigation progress to provide contextual and progressive guidance without directly revealing solutions. The platform also implements a discovery-based investigation framework that allows learners to solve investigation objectives using multiple valid approaches rather than following a single predefined path.

The system was developed using React.js for the frontend, Node.js and Express.js for backend services, PostgreSQL for data management, and xterm.js for terminal simulation. Three investigation scenarios were implemented, namely Brute Force Login Investigation, Suspicious Script Execution Investigation, and SSH Forensics Investigation. The completed platform provides a safe, accessible, and engaging environment for developing foundational cybersecurity investigation skills while reducing the complexity and infrastructure requirements commonly associated with traditional cybersecurity training environments.

ABSTRAK

[ABSTRAK AKAN DITAMBAH KEMUDIAN]

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
DB	Database
ERD	Entity Relationship Diagram
JWT	JSON Web Token
LLM	Large Language Model
PTY	Pseudo Terminal
RCE	Remote Code Execution
SPA	Single Page Application
SQL	Structured Query Language
UI	User Interface
UX	User Experience
VM	Virtual Machine
XP	Experience Points

CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid advancement of digital technologies, cloud computing, and interconnected systems has significantly increased the importance of cybersecurity across modern organizations and industries. As cyber threats continue to evolve in both complexity and frequency, there is a growing demand for cybersecurity professionals who possess not only theoretical knowledge but also practical investigation and analytical skills. Educational institutions therefore play a critical role in preparing students with the competencies required to identify, analyze, and respond to cybersecurity incidents effectively.

Among the essential skills required in defensive cybersecurity are log analysis, system investigation, and digital evidence discovery. These skills enable security analysts to reconstruct attack timelines, identify suspicious activities, determine the root cause of incidents, and support incident response processes. However, developing such competencies requires more than traditional classroom instruction. Learners must be exposed to practical investigation activities that allow them to interact with realistic scenarios and apply cybersecurity concepts in simulated environments.

Despite the increasing importance of practical cybersecurity education, many undergraduate Information Technology (IT) and software students continue to experience limited exposure to realistic investigative activities. Traditional learning approaches often emphasize theoretical concepts through lectures, reading materials, and written assessments while providing limited opportunities for learners to practice log analysis and system investigation tasks. As a result, students may graduate with a strong understanding of cybersecurity concepts but insufficient practical experience in applying investigative techniques within realistic environments.

Furthermore, existing cybersecurity training environments often rely on virtual machines, cyber ranges, or containerized infrastructures to provide practical learning

experiences. While these environments offer realistic simulations, they may require substantial hardware resources, infrastructure management, and technical setup. Such complexity can present challenges for educational institutions and may create barriers for beginner learners who are unfamiliar with advanced cybersecurity environments.

In recent years, gamification and simulation-based learning have gained increasing attention as effective approaches for improving learner engagement and participation. Gamification integrates elements such as achievements, progression systems, scoring mechanisms, and challenges into educational activities to create a more motivating learning experience. At the same time, simulation-based learning allows students to practice skills within controlled environments that mimic real-world situations without exposing actual systems to security risks.

In addition, advancements in Artificial Intelligence (AI) have enabled the development of intelligent learning support systems that can assist learners during educational activities. Rather than providing direct solutions, AI-assisted guidance systems can analyze learner behaviour and provide contextual hints that encourage independent problem solving while reducing frustration during complex tasks.

To address these challenges, this project develops KINETIC BREACH, a web-based gamified training platform designed to improve log analysis and system investigation skills among undergraduate students. The platform provides a simulated Linux-like investigation environment where learners can interact with a virtual filesystem, examine logs, analyze system artefacts, and investigate cybersecurity incidents using command-line tools within a browser-based environment. The system also incorporates gamification features, adaptive AI-assisted guidance, and investigation-based learning activities to create a more engaging and effective educational experience.

Overall, KINETIC BREACH aims to bridge the gap between theoretical cybersecurity education and practical investigation skill development by providing a safe, accessible, and interactive platform that supports hands-on learning and encourages analytical thinking through realistic cybersecurity investigation scenarios.

1.2 Problem Statement

Despite the growing demand for cybersecurity professionals, several challenges continue to affect the effectiveness of cybersecurity education, particularly in the development of practical investigation skills among undergraduate students. Based on the review of existing educational approaches and cybersecurity training environments, the following problems have been identified.

PS1: Lack of Practical Investigation Training

Many undergraduate Information Technology (IT) and software development programmes focus primarily on theoretical cybersecurity concepts and predefined laboratory exercises. Although these approaches provide fundamental knowledge, they often offer limited opportunities for students to perform realistic investigation activities such as log analysis, system examination, evidence discovery, and incident reconstruction. Consequently, learners may have insufficient practical experience in applying investigative techniques within simulated cybersecurity environments.

PS2: Passive Learning and Low Learner Engagement

Traditional cybersecurity learning methods commonly rely on lectures, reading materials, video-based content, and written assessments. While these approaches are effective for knowledge delivery, they often provide limited interaction and active participation opportunities. As a result, students may experience reduced motivation, engagement, and interest during the learning process, which can negatively affect knowledge retention and practical skill development.

PS3: Complexity and Accessibility Issues of Existing Cybersecurity Training Environments

Many practical cybersecurity training platforms rely on virtual machines, cyber ranges, or container-based infrastructures to simulate realistic environments. Although these platforms provide valuable hands-on experiences, they often require significant hardware resources, technical setup, and infrastructure management. Such complexity may create accessibility challenges for educational institutions and beginner learners who lack prior experience with advanced cybersecurity environments.

PS4: Limited Learning Support During Investigation Activities

Investigation-based cybersecurity exercises often require learners to analyze large amounts of information, explore multiple evidence sources, and identify relevant findings independently. When learners become stuck during an investigation, existing platforms may provide either insufficient assistance or overly detailed walkthroughs that reveal solutions directly. This can reduce learner confidence, hinder independent problem-solving, and negatively affect the overall learning experience.

Therefore, there is a need for an interactive and accessible cybersecurity training platform that supports practical investigation activities, promotes learner engagement, provides contextual learning assistance, and enables students to develop foundational log analysis and system investigation skills through realistic simulation-based learning

experiences.

1.3 Objectives

The objectives of this project are as follows:

1. To design realistic cybersecurity investigation scenarios that enable learners to practice log analysis and system investigation skills in a simulated environment.
2. To develop a web-based gamified training platform that promotes active learning and engagement through interactive investigation activities.
3. To implement a virtual terminal environment and database-driven virtual filesystem that provide a safe and accessible alternative to traditional cybersecurity training infrastructures.
4. To implement an AI-assisted guidance system that provides contextual hints to support learners during investigation activities without directly revealing solutions.
5. To incorporate gamification elements such as scoring, progression systems, achievements, and performance tracking to enhance learner motivation and engagement.
6. To evaluate the effectiveness and usability of the developed platform in supporting cybersecurity investigation skill development among undergraduate IT and software students through user testing and structured feedback.

1.4 Scope

The scope of this project defines the boundaries, functionalities, target users, development environment, and limitations of the developed system. Establishing a clear project scope ensures that the project remains focused on achieving its objectives while maintaining feasibility within the allocated development period. The scope of KINETIC BREACH is divided into four main areas: targeted users, system modules, development environment, and project limitations.

1.4.1 Targeted Users

This section identifies the intended users of the developed system. KINETIC BREACH is designed to support learners who require practical experience in log analysis and system investigation within a safe and controlled training environment.

The targeted users of the system are as follows:

- **Undergraduate Information Technology and Software Development Students**

  The primary target users are undergraduate students enrolled in Information Technology, Software Development, and related computing programmes. These students often possess theoretical cybersecurity knowledge but may have limited opportunities to practice investigation techniques within realistic environments.

- **Cybersecurity Students**

  Students specializing in cybersecurity can utilize the platform to strengthen their log analysis, system investigation, and evidence discovery skills through interactive investigation scenarios.

- **Beginner Learners and Technology Enthusiasts**

  The platform is also suitable for beginner learners who wish to gain hands-on experience with Linux-based investigation techniques without requiring advanced technical knowledge or complex laboratory setups.

1.4.2 System Modules

This section describes the major functional modules implemented within KINETIC BREACH. These modules work together to provide an interactive and engaging investigation-based learning environment.

- **Authentication and User Management Module**

  This module manages user registration, authentication, profile management, and user progression data. It ensures that users can securely access the platform and maintain their learning progress.

- **Scenario Management Module**

  This module manages the cybersecurity investigation scenarios available within the system. It stores scenario information, mission briefings, objectives, and difficulty levels that guide learners throughout each investigation.

- **Virtual Terminal and Filesystem Simulation Module**

  This module provides a browser-based Linux-like terminal environment supported by a database-driven virtual filesystem. Users can execute investigation commands and interact with simulated system artefacts without accessing a real operating system.

- **Investigation and Discovery Module**

  This module evaluates user actions and tracks investigation progress. It allows learners to discover evidence, complete objectives, and progress through investigation activities using multiple valid approaches.

- **AI-Assisted Guidance Module**

  This module provides contextual hints and guidance when learners encounter difficulties during investigation activities. The guidance mechanism is designed to support learning without directly revealing solutions.

- **Gamification and Progress Tracking Module**

  This module incorporates gamification features including scoring mechanisms, achievements, progression systems, experience points (XP), and performance tracking to encourage learner engagement and motivation.

1.4.3 Development Environment

This section describes the technologies and tools used to develop the system.

- Frontend Development: React.js
- Backend Development: Node.js and Express.js
- Database Management System: PostgreSQL
- Terminal Interface: xterm.js
- Artificial Intelligence Integration: OpenAI API
- Version Control System: Git and GitHub
- Development Tools: Visual Studio Code

These technologies were selected to provide a scalable, maintainable, and interactive web-based learning platform while supporting efficient system development and deployment.

1.4.4 Limitations

Although KINETIC BREACH provides an effective platform for cybersecurity investigation training, several limitations are defined within the scope of the project.

- The system currently provides three investigation scenarios: Brute Force Login Investigation, Suspicious Script Execution Investigation, and SSH Forensics Investigation.
- The platform focuses exclusively on Linux-based investigation activities and does not cover Windows or mobile device forensics.
- All investigations are performed within a simulated environment and do not involve real operating systems, virtual machines, or live network infrastructures.
- The platform focuses on defensive cybersecurity investigation skills and does not include penetration testing, offensive security exercises, or real-world attack execution.
- Advanced shell functionalities beyond the supported command set are outside the scope of the current implementation.

1.5 Project Significance

The development of KINETIC BREACH provides several benefits to students, educators, and institutions involved in cybersecurity education. By combining investigation-based learning, gamification, and AI-assisted guidance within a web-based environment, the platform contributes to improving both the accessibility and effectiveness of practical cybersecurity training.

1.5.1 Benefits to Students

The platform provides students with opportunities to develop practical log analysis and system investigation skills through realistic cybersecurity scenarios. Unlike traditional learning methods that focus primarily on theory, KINETIC BREACH allows learners to apply their knowledge in a hands-on environment. The simulated terminal and investigation activities help students gain experience in evidence discovery, system analysis, and incident investigation while reducing the risks associated with real-world environments.

1.5.2 Benefits to Educators

The platform can serve as an educational tool that supports practical teaching activities related to cybersecurity and digital forensics. Educators may utilize the investigation scenarios to supplement classroom instruction and provide students with structured practical exercises. In addition, the scoring and progress tracking mechanisms can assist instructors in monitoring student participation and performance throughout the learning process.

1.5.3 Benefits to Educational Institutions

Educational institutions can benefit from a cost-effective alternative to traditional cybersecurity laboratories that often require dedicated hardware, virtual machines, or complex infrastructure management. The web-based nature of the platform enables practical cybersecurity training to be delivered through standard web browsers, reducing deployment complexity and improving accessibility for learners.

1.5.4 Benefits to Cybersecurity Education

This project contributes to the advancement of cybersecurity education by demonstrating an alternative approach to practical investigation training. The integration of gamification, investigation-based learning, and AI-assisted guidance provides a framework that may improve learner engagement and encourage independent problem-solving. Furthermore, the project highlights the potential of simulated environments as accessible training solutions for developing foundational cybersecurity investigation skills.

Overall, KINETIC BREACH aims to bridge the gap between theoretical cybersecurity knowledge and practical investigation experience by providing an engaging, accessible, and educational learning platform.

1.6 Expected Output

The expected output of this project is a fully functional web-based cybersecurity training platform known as KINETIC BREACH. The platform is designed to provide learners with an interactive environment for developing log analysis and system investigation skills through realistic cybersecurity investigation scenarios.

The primary outputs of the project are as follows:

1. **Web-Based Gamified Training Platform**

   A complete web application that allows users to access cybersecurity investigation activities through a standard web browser without requiring additional software installations or specialized laboratory environments.

2. **Virtual Terminal and Filesystem Simulation Environment**

   A browser-based Linux-like terminal environment supported by a database-driven virtual filesystem. This environment enables users to perform investigation activities using command-line tools within a safe and controlled simulation.

3. **Cybersecurity Investigation Scenarios**

   Three interactive investigation scenarios are implemented within the platform:

   - Brute Force Login Investigation
   - Suspicious Script Execution Investigation
   - SSH Forensics Investigation

   These scenarios provide learners with practical experience in log analysis, evidence discovery, and system investigation activities.

4. **AI-Assisted Guidance System**

   An intelligent guidance component that monitors learner progress and provides contextual hints when users encounter difficulties during investigation activities. The guidance system is designed to encourage independent problem-solving without directly revealing solutions.

5. **Gamification and Progress Tracking Features**

   A gamification framework that includes scoring mechanisms, experience points (XP), achievements, rankings, and performance tracking to improve learner engagement and motivation throughout the learning process.

6. **Performance Evaluation and Feedback**

   A performance evaluation mechanism that assesses learner progress based on investigation activities, objective completion, and overall scenario performance. The generated results provide feedback that supports skill development and self-assessment.

Overall, the completed project is expected to provide an accessible, engaging, and practical cybersecurity training platform that supports the development of investigation skills through simulation-based learning.

1.7 Conclusion

This chapter introduced the background, motivation, objectives, scope, significance, and expected outputs of KINETIC BREACH, a gamified training platform designed to improve log analysis and system investigation skills among undergraduate students. The chapter highlighted the challenges associated with practical cybersecurity education, including limited hands-on investigation opportunities, low learner engagement, the complexity of traditional training environments, and the need for effective learning support during investigation activities.

To address these challenges, the project proposes a web-based platform that combines investigation-based learning, virtual terminal simulation, AI-assisted guidance, and gamification mechanisms within a controlled environment. The project objectives, scope, significance, and expected outcomes were also presented to establish the foundation and direction of the system development process.

The next chapter presents the literature review and project methodology. It discusses relevant concepts, existing systems, technologies, and approaches related to cybersecurity training, gamification, investigation-based learning, and AI-assisted educational support. In addition, the chapter describes the methodology adopted for the development of KINETIC BREACH.

CHAPTER 2

LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

The successful development of an educational cybersecurity platform requires a strong understanding of the concepts, technologies, learning approaches, and development methodologies that support its implementation. Therefore, before designing and developing KINETIC BREACH, it is important to examine the existing knowledge, educational practices, and technical solutions related to cybersecurity training and investigation-based learning.

Cybersecurity education has evolved significantly in recent years due to the increasing demand for professionals capable of responding to modern security incidents. As organizations face increasingly sophisticated cyber threats, educational institutions are expected to provide learners with not only theoretical knowledge but also practical investigation and analytical skills. Consequently, various educational approaches such as gamification, interactive learning, simulation-based training, and intelligent learning support systems have emerged to enhance learner engagement and practical skill development.

In addition, numerous cybersecurity training platforms have been developed to provide hands-on learning experiences through challenge-based activities, virtual laboratories, and investigation-oriented exercises. Studying these existing systems is important for identifying their strengths, limitations, and potential areas for improvement that can be addressed by the proposed system.

This chapter presents the facts, findings, and theoretical foundations relevant to the development of KINETIC BREACH. It reviews concepts related to cybersecurity education, gamification, interactive learning, and simulation-based training environments. Existing cybersecurity learning platforms are also examined and compared to

identify the research and implementation gap addressed by the proposed system.

Furthermore, this chapter discusses the selected development methodology, project requirements, and project schedule adopted throughout the implementation of KINETIC BREACH. These discussions establish the theoretical, technical, and methodological foundation that guides the design and development of the proposed system.

2.2 Facts and Findings

This section presents the theoretical foundations, existing technologies, and relevant educational approaches that support the development of KINETIC BREACH. The purpose of this review is to identify current practices, challenges, and opportunities within cybersecurity education and practical investigation training.

The reviewed materials focus on domains related to cybersecurity education, gamification, interactive learning, simulation-based training environments, artificial intelligence assisted learning support, and digital forensics investigation. Existing cybersecurity training platforms are also examined to identify their strengths, limitations, and relevance to the proposed system. Finally, several implementation techniques and training approaches are analyzed to justify the selection of the methodology and technologies adopted in KINETIC BREACH.

The findings presented in this section provide the theoretical and technical foundation for the design and development decisions discussed in subsequent chapters.

2.2.1 Domain

This section reviews the key concepts, educational approaches, and technological domains relevant to the development of KINETIC BREACH. Understanding these domains is essential for identifying the challenges faced in cybersecurity education and the methods commonly used to improve practical skill development. The reviewed domains include cybersecurity education and training, gamification, interactive learning, simulation-based training, artificial intelligence assisted learning support, and digital forensics investigation.

2.2.1.1 Cybersecurity Education and Training

The rapid growth of digital technologies, cloud computing, and interconnected systems has significantly increased the importance of cybersecurity across various in-

dustries. Organizations today rely heavily on digital infrastructures to support business operations, communication, and data management. As a result, cybersecurity threats such as unauthorized access, malware infections, credential theft, and data breaches have become increasingly common. This growing threat landscape has created a strong demand for cybersecurity professionals capable of protecting systems and responding to security incidents effectively.

To meet this demand, educational institutions are expected to equip students with the knowledge and skills required to operate within modern cybersecurity environments. Traditional cybersecurity education often focuses on theoretical concepts such as network security, cryptography, operating systems, and information assurance. While these subjects provide essential foundational knowledge, they may not adequately prepare students for the practical challenges encountered in real-world cybersecurity roles.

Several studies have highlighted the gap between academic preparation and industry expectations. Many graduates possess a strong understanding of cybersecurity principles but have limited experience applying investigative techniques within realistic environments. Tasks such as analyzing authentication logs, identifying suspicious activities, examining system artefacts, and reconstructing incident timelines require practical experience that cannot be fully developed through lectures and written assessments alone.

In response to these challenges, cybersecurity education has increasingly shifted toward experiential and learner-centered approaches that emphasize practical skill development. Educational institutions are adopting laboratory exercises, simulation environments, cybersecurity competitions, and investigation-based learning activities to provide learners with opportunities to apply theoretical knowledge in realistic scenarios. These approaches encourage active participation and help students develop analytical thinking, problem-solving abilities, and practical investigation skills.

Furthermore, modern cybersecurity training increasingly emphasizes defensive cybersecurity competencies, including digital forensics, log analysis, incident investigation, and security monitoring. These skills are essential for cybersecurity professionals responsible for identifying, investigating, and responding to security incidents within organizational environments.

Overall, the literature indicates that cybersecurity education is evolving from traditional knowledge-based instruction toward more practical and interactive learning approaches. This evolution aims to better prepare learners for industry requirements by combining theoretical understanding with hands-on experience and investigation-based skill development.

2.2.1.2 Gamification in Education

Gamification refers to the application of game-design elements and mechanics within non-game environments to enhance user engagement, motivation, and participation. Common gamification elements include points, achievements, badges, levels, rankings, rewards, challenges, and progression systems. In educational settings, these elements are integrated into learning activities to create more engaging and interactive learning experiences.

In recent years, gamification has gained significant attention within the education sector due to its potential to improve learner motivation and participation. Traditional learning environments often rely on lectures, reading materials, and written assessments, which may result in reduced learner engagement, particularly when dealing with complex or highly technical subjects. Gamification addresses this challenge by introducing elements that encourage active involvement and continuous participation throughout the learning process.

Research suggests that gamified learning environments can positively influence learner behaviour by increasing motivation, improving knowledge retention, and encouraging the completion of educational tasks. Through progression-based systems, learners receive immediate feedback regarding their performance and achievements, allowing them to monitor their progress and maintain interest in learning activities. This sense of accomplishment can contribute to higher levels of engagement and persistence when facing challenging tasks.

Within cybersecurity education, gamification has been widely adopted through platforms that incorporate challenge-based learning, capture-the-flag competitions, and scenario-based exercises. These environments encourage learners to actively solve problems, investigate evidence, and apply cybersecurity concepts through practical interaction rather than passive observation. As a result, learners are more likely to develop analytical thinking and problem-solving skills while remaining engaged in the learning process.

Furthermore, gamification can reduce learner frustration during complex technical activities by providing structured progression paths, achievement systems, and performance feedback. Such mechanisms help learners remain motivated while gradually developing confidence and competency in performing practical cybersecurity tasks.

Overall, gamification has emerged as an effective educational approach that supports learner engagement, motivation, and practical skill development. Its ability to transform learning activities into interactive and rewarding experiences makes it

particularly suitable for cybersecurity training environments where active participation and continuous practice are essential for skill acquisition.

2.2.1.3 Interactive and Hands-on Learning

Interactive learning refers to educational approaches that actively involve learners in the learning process through participation, exploration, problem-solving, and practical activities. Unlike traditional teaching methods that primarily focus on one-way knowledge delivery, interactive learning encourages learners to engage directly with educational content and apply concepts through active involvement.

Educational research consistently demonstrates that active participation enhances learner engagement, knowledge retention, and overall learning outcomes. When learners are required to analyze information, make decisions, and solve problems independently, they develop a deeper understanding of the subject matter compared to passive learning approaches. Interactive learning environments also encourage critical thinking and promote the development of practical skills that can be applied beyond the classroom.

Hands-on learning is a specific form of interactive learning that emphasizes direct experience and practical application. Rather than simply studying concepts theoretically, learners perform tasks, interact with tools, and gain experience through experimentation and practice. This approach allows students to learn by doing, which can improve confidence and strengthen the connection between theoretical knowledge and practical implementation.

In cybersecurity education, hands-on learning is particularly important because many cybersecurity tasks involve investigation, analysis, and decision-making processes that cannot be fully understood through theoretical instruction alone. Activities such as examining logs, identifying suspicious system behaviour, tracing digital evidence, and reconstructing security incidents require learners to actively engage with information and apply investigative techniques in realistic situations.

Furthermore, hands-on learning environments provide opportunities for learners to make mistakes and learn from them within safe and controlled settings. This experiential learning process encourages exploration and independent problem-solving while reducing the risks associated with real-world systems. Such environments are especially valuable for beginner learners who may lack prior practical experience in cybersecurity investigations.

As a result, many modern cybersecurity educational platforms increasingly incorporate interactive exercises, investigation-based activities, and practical simulations

to improve learner engagement and support skill development. These approaches help bridge the gap between theoretical understanding and practical competency, enabling learners to develop the investigative and analytical skills required in cybersecurity-related roles.

Overall, interactive and hands-on learning approaches play a critical role in cybersecurity education by providing learners with meaningful opportunities to apply knowledge, develop practical skills, and gain confidence through active participation and experiential learning.

2.2.1.4 Simulation-Based Cybersecurity Training

Simulation-based learning has become an increasingly important component of modern cybersecurity education. This educational approach allows learners to practice cybersecurity tasks and investigative procedures within controlled environments that imitate real-world systems and security incidents. By providing realistic scenarios without exposing actual systems to operational risks, simulation-based training enables learners to develop practical skills in a safe and structured manner.

Cybersecurity simulation environments are commonly designed to replicate system behaviours, network activities, security events, and incident response situations. Within these environments, learners can interact with simulated artefacts, analyze system logs, investigate suspicious activities, and apply cybersecurity concepts to realistic situations. Such practical experiences help learners bridge the gap between theoretical knowledge and real-world application.

One of the primary advantages of simulation-based training is its ability to provide hands-on learning opportunities without the risks associated with real operational environments. Learners can repeatedly practice investigation techniques, experiment with different approaches, and learn from mistakes without affecting actual systems or infrastructures. This flexibility makes simulation-based learning particularly valuable for beginner learners who require structured environments to build confidence and practical experience.

Various simulation-based cybersecurity training solutions have been developed in recent years, including Cyber Ranges, virtual laboratories, and challenge-based platforms. These systems provide learners with opportunities to experience realistic cybersecurity scenarios while developing technical and analytical skills. Many of these environments support activities such as incident response, digital forensics, vulnerability analysis, and system investigation.

Despite their educational benefits, traditional simulation environments often

require substantial hardware resources, infrastructure management, and technical configuration. Virtual machines, containerized environments, and large-scale cyber ranges may introduce complexity that can increase deployment costs and create accessibility challenges for educational institutions and beginner learners. Additionally, the maintenance and administration of such environments may require significant technical expertise.

As a result, there is increasing interest in alternative simulation approaches that can provide practical learning experiences while reducing infrastructure complexity. Browser-based and lightweight simulation environments offer a promising solution by enabling learners to perform investigation activities without requiring dedicated laboratory setups or advanced technical configurations.

Overall, simulation-based cybersecurity training plays a critical role in developing practical cybersecurity competencies. By providing realistic, safe, and interactive learning environments, simulation-based approaches help learners acquire the investigative and analytical skills necessary for modern cybersecurity practice.

2.2.1.5 Artificial Intelligence Assisted Learning Support

Artificial Intelligence (AI) has become increasingly integrated into modern educational systems due to its ability to provide adaptive and personalized learning experiences. AI-powered educational technologies can analyze learner behaviour, monitor progress, and deliver contextual support that helps learners overcome challenges encountered during the learning process. As a result, AI has emerged as a valuable tool for enhancing both learner engagement and educational effectiveness.

Traditional educational environments typically provide the same learning materials and instructional support to all learners regardless of their individual needs or learning progress. However, learners often possess different levels of prior knowledge, learning speed, and problem-solving abilities. AI-assisted learning systems address this challenge by adapting educational support based on learner performance and behaviour, allowing more personalized learning experiences.

One common application of AI in education is the provision of intelligent guidance and contextual feedback. Rather than directly providing answers, AI systems can offer hints, recommendations, and learning assistance that encourage learners to think critically and solve problems independently. This approach supports active learning while reducing learner frustration when complex tasks are encountered.

In practical and technical disciplines such as cybersecurity, AI-assisted support can be particularly beneficial. Learners are frequently required to analyze large

amounts of information, investigate unfamiliar systems, and identify relevant evidence during practical exercises. Intelligent support mechanisms can assist learners by providing contextual guidance and helping them maintain progress without removing the investigative and exploratory nature of the learning activity.

Recent advancements in large language models (LLMs) and intelligent educational systems have further expanded the capabilities of AI-assisted learning environments. These technologies can interpret learner interactions, identify potential difficulties, and generate adaptive responses that support learning objectives while maintaining user engagement.

Overall, AI-assisted learning support represents a promising approach for enhancing educational experiences by providing adaptive guidance, reducing learner frustration, and encouraging independent problem-solving. The integration of such capabilities within practical training environments can contribute significantly to learner engagement and skill development, particularly in complex technical domains such as cybersecurity.

2.2.1.6 Digital Forensics and Log Analysis Training

Digital forensics and log analysis are fundamental components of defensive cybersecurity and incident response activities. These disciplines focus on the collection, examination, and interpretation of digital evidence in order to identify security incidents, understand system behaviour, and reconstruct the sequence of events that occurred during an attack or suspicious activity.

System logs serve as one of the most important sources of digital evidence in cybersecurity investigations. Operating systems, applications, authentication services, and network devices continuously generate logs that record system activities and user actions. Security analysts rely on these records to identify abnormal behaviour, detect security incidents, trace attacker activities, and support incident response processes.

Log analysis involves examining and interpreting recorded events to identify meaningful patterns and security-relevant information. Common investigation activities include reviewing authentication records, identifying failed login attempts, tracing unauthorized access attempts, examining command histories, and analyzing system events. These activities enable investigators to determine what happened, when it occurred, and how the incident affected the target system.

Digital forensics extends beyond log examination by incorporating evidence collection, artefact analysis, and timeline reconstruction. Investigators often examine files, user activities, system configurations, and other digital artefacts to establish

relationships between events and identify the root causes of incidents. This investigative process requires analytical thinking, attention to detail, and the ability to correlate information obtained from multiple sources.

The growing importance of cybersecurity incident response has increased the demand for professionals who possess practical digital investigation skills. Consequently, educational institutions are placing greater emphasis on providing learners with opportunities to develop competencies related to evidence discovery, log analysis, and incident investigation. However, acquiring these skills can be challenging because effective training typically requires realistic investigation scenarios and access to representative system artefacts.

Simulation-based investigation environments have emerged as an effective solution for addressing this challenge. By providing learners with realistic logs, files, and investigation objectives within controlled environments, these systems enable students to practice investigative procedures without requiring access to real production systems. Learners can analyze evidence, identify suspicious activities, and develop investigation strategies while gaining practical experience in a safe educational setting.

Overall, digital forensics and log analysis training play a critical role in preparing learners for defensive cybersecurity roles. The ability to analyze logs, interpret evidence, and reconstruct security incidents represents an essential competency for cybersecurity professionals responsible for monitoring, investigating, and responding to security threats within modern digital environments.

2.2.2 Existing Systems

Various cybersecurity training platforms have been developed to provide learners with practical cybersecurity experience through interactive exercises, simulation environments, and challenge-based activities. These platforms aim to improve technical skills, problem-solving abilities, and learner engagement by providing opportunities for hands-on practice within controlled environments.

Studying existing systems is important for identifying current approaches, strengths, limitations, and educational features commonly adopted within cybersecurity training platforms. The findings obtained from this review help establish the design considerations and implementation decisions adopted in the development of KINETIC BREACH.

The following systems were selected for review because they are widely recognized within the cybersecurity education community and provide practical learning experiences relevant to the objectives of this project.

2.2.2.1 Hack The Box

Hack The Box (HTB) is a widely recognized cybersecurity training platform that provides realistic and hands-on learning environments for cybersecurity enthusiasts, students, and professionals. The platform offers a large collection of virtual machines, laboratories, and challenge-based exercises designed to simulate real-world cybersecurity scenarios.

Hack The Box primarily focuses on practical cybersecurity skills through activities such as penetration testing, vulnerability assessment, privilege escalation, digital forensics, and incident investigation. Learners are required to interact directly with target systems, identify vulnerabilities, and solve technical challenges using various cybersecurity tools and techniques.

One of the major strengths of Hack The Box is its highly realistic training environment. The platform provides users with access to systems that closely resemble real-world infrastructures, allowing learners to gain valuable practical experience. In addition, the platform contains a wide variety of challenges that accommodate different skill levels, ranging from beginner to advanced cybersecurity professionals.

However, despite its educational value, Hack The Box may present challenges for beginner learners. Many exercises assume prior cybersecurity knowledge and familiarity with Linux systems, networking concepts, and penetration testing methodologies. As a result, new learners may find the platform difficult to navigate without additional guidance or structured learning support.

Furthermore, the platform primarily emphasizes offensive cybersecurity activities such as exploitation and penetration testing. Although investigation and forensic-related challenges are available, they are not the primary focus of the learning experience. This may limit its suitability for learners specifically seeking structured training in log analysis and system investigation skills.

Strengths

- Highly realistic cybersecurity environments.
- Extensive collection of practical challenges and laboratories.
- Strong emphasis on hands-on learning and technical skill development.
- Suitable for intermediate and advanced learners.

Limitations

- May be challenging for beginners with limited cybersecurity knowledge.

- Limited adaptive learning support and guided assistance.
- Primarily focused on offensive cybersecurity activities.
- Requires interaction with more complex training infrastructures compared to lightweight educational platforms.

2.2.2.2 TryHackMe

TryHackMe is an online cybersecurity learning platform designed to provide structured and beginner-friendly cybersecurity education through interactive and guided learning experiences. The platform combines theoretical explanations, practical exercises, and hands-on activities to help learners develop foundational cybersecurity knowledge and technical skills.

Unlike many cybersecurity platforms that focus primarily on advanced technical challenges, TryHackMe emphasizes accessibility and progressive learning. The platform organizes educational content into learning paths that guide users through various cybersecurity topics, including networking, operating systems, web security, digital forensics, penetration testing, and incident response. This structured approach makes the platform particularly suitable for students and individuals with limited prior cybersecurity experience.

One of the key strengths of TryHackMe is its ability to combine practical activities with guided instruction. Learners are provided with clear objectives, supporting explanations, and step-by-step exercises that help them understand technical concepts while developing practical skills. This educational approach reduces the learning curve and allows beginners to gain confidence when interacting with cybersecurity tools and environments.

The platform also incorporates gamification elements such as progress tracking, completion badges, learning streaks, and achievement systems. These features help maintain learner motivation and encourage continuous participation throughout the learning process.

However, despite its educational effectiveness, many TryHackMe learning rooms rely heavily on guided walkthroughs and predefined instructions. While this approach supports beginner learners, it may reduce opportunities for independent investigation and exploratory problem-solving. Learners are often directed toward specific solutions rather than discovering evidence and investigation paths through their own analytical processes.

Additionally, although the platform includes digital forensics and incident re-

sponse content, much of the learning experience remains focused on completing guided tasks rather than conducting open-ended investigations. As a result, learners may have fewer opportunities to develop independent log analysis and system investigation skills compared to investigation-focused simulation environments.

Strengths

- Beginner-friendly learning environment.
- Structured learning paths and guided exercises.
- Strong educational support and accessibility.
- Includes gamification and progress tracking features.

Limitations

- Many activities rely heavily on guided walkthroughs.
- Limited emphasis on independent investigation and evidence discovery.
- Less focus on open-ended system investigation workflows.
- Adaptive contextual guidance is limited.

2.2.2.3 OverTheWire

OverTheWire is a cybersecurity learning platform that provides a collection of wargames designed to teach Linux, networking, and security concepts through command-line interaction. The platform is widely recognized for its terminal-focused learning approach, where users solve progressively challenging tasks by connecting to remote systems and executing commands within Linux environments.

Unlike many modern cybersecurity training platforms that rely on graphical interfaces, OverTheWire emphasizes direct interaction with the command line. Learners are required to navigate file systems, analyze files, manipulate permissions, and utilize various Linux utilities to complete objectives. This approach helps users develop familiarity with Linux commands and problem-solving techniques that are commonly used in cybersecurity-related activities.

One of the major strengths of OverTheWire is its simplicity and accessibility. The platform requires minimal system resources and focuses heavily on practical command-line experience. Learners gain valuable exposure to Linux environments

while developing independent learning and analytical skills through exploration and experimentation.

However, OverTheWire provides limited educational support and guidance. Learners are generally expected to discover solutions independently, which may present challenges for beginners with limited prior knowledge. In addition, the platform includes minimal gamification features and lacks adaptive learning mechanisms that can provide contextual assistance when learners encounter difficulties.

Furthermore, although OverTheWire successfully develops command-line proficiency, it places less emphasis on realistic cybersecurity investigations, log analysis activities, and structured incident response workflows. As a result, learners may gain strong technical command-line skills but receive limited exposure to investigation-oriented cybersecurity scenarios.

Strengths

- Strong focus on command-line and Linux skill development.
- Encourages independent problem-solving and exploration.
- Lightweight and accessible learning environment.
- Provides practical hands-on learning experiences.

Limitations

- Limited gamification and progression features.
- Minimal learner guidance and educational support.
- Lacks adaptive assistance mechanisms.
- Limited focus on realistic investigation and log analysis activities.

2.2.2.4 PicoCTF

PicoCTF is an educational cybersecurity competition platform developed to introduce students and beginners to cybersecurity concepts through challenge-based learning activities. The platform provides a large collection of cybersecurity challenges covering areas such as cryptography, web security, reverse engineering, digital forensics, and binary exploitation.

The primary objective of PicoCTF is to make cybersecurity education more engaging and accessible through competition-oriented activities. Learners solve challenges, earn points, and progress through increasingly difficult tasks while developing technical problem-solving skills. The platform is widely used in educational institutions due to its beginner-friendly design and strong emphasis on learning through exploration.

One of the key strengths of PicoCTF is its ability to make cybersecurity concepts approachable for new learners. The challenge-based structure encourages active participation and curiosity while allowing students to learn at their own pace. Additionally, the scoring and competition systems help maintain learner motivation and engagement throughout the learning experience.

Despite its educational value, PicoCTF primarily focuses on individual challenge completion rather than realistic investigation workflows. Many activities are designed as isolated tasks that emphasize solving specific technical problems rather than conducting comprehensive cybersecurity investigations. Consequently, learners may have limited opportunities to practice incident analysis, evidence correlation, and systematic investigation procedures.

Furthermore, the platform does not provide adaptive guidance systems that can analyze learner behaviour and offer personalized assistance. While hints may be available for certain challenges, the support mechanisms are generally static and do not adapt dynamically to learner progress.

Strengths

- Beginner-friendly cybersecurity learning platform.
- Engaging challenge-based educational approach.
- Strong gamification and scoring mechanisms.
- Encourages problem-solving and technical exploration.

Limitations

- Primarily challenge-oriented rather than investigation-oriented.
- Limited emphasis on realistic cybersecurity investigation workflows.
- Minimal adaptive learning support.
- Activities are often isolated rather than scenario-based.

2.2.2.5 Comparison of Existing Systems

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Table 2.1 presents a comparison between the reviewed cybersecurity training platforms and the proposed system.

Table 2.1: Comparison of Existing Cybersecurity Training Platforms

Feature	**KINETIC BREACH**	**HTB**	**THM**	**OTW**	**PicoCTF**
Beginner-Friendly Learning	Yes	Partial	Yes	Partial	Yes
Investigation-Based Learning	Yes	Partial	Limited	Limited	Limited
Terminal-Based Interaction	Yes	Yes	Partial	Yes	Partial
Gamification Features	Yes	Yes	Yes	Limited	Yes
AI-Assisted Guidance	Yes	No	No	No	No
Simulation-Based Investigation	Yes	Yes	Partial	Limited	Limited
Lightweight Educational Environment	Yes	No	Partial	Yes	Yes
Focus on Log and System Investigation	Yes	Partial	Limited	Limited	Limited

Based on the comparison, existing cybersecurity training platforms provide valuable educational experiences but often focus on challenge completion, offensive security activities, or guided learning paths. In contrast, KINETIC BREACH emphasizes log analysis, system investigation, evidence discovery, and adaptive learner support within a lightweight browser-based simulation environment. The proposed system combines gamification, investigation-based learning, and AI-assisted guidance to provide a more focused learning experience for developing foundational cybersecurity investigation skills.

2.2.3 Technique

Various techniques and implementation approaches can be adopted when developing cybersecurity training environments. Each approach offers different advantages and limitations in terms of realism, infrastructure requirements, accessibility, scalability, and educational effectiveness. Therefore, it is important to analyze existing approaches before selecting the most suitable technique for the proposed system.

The following subsections discuss several commonly used cybersecurity training approaches and justify the selected technique adopted in the development of KINETIC BREACH.

2.2.3.1 Virtual Machine-Based Training

Virtual Machine (VM)-based training environments are widely used in cybersecurity education because they provide highly realistic operating system environments for learners. In this approach, learners interact with complete operating systems running inside virtual machines that simulate real-world systems and infrastructures.

VM-based environments allow learners to perform various cybersecurity activities, including system administration, incident response, penetration testing, and digital forensics. Since learners interact with actual operating systems, these environments provide a high level of realism and closely resemble professional cybersecurity workplaces.

However, VM-based environments require significant hardware resources, storage capacity, and infrastructure management. Educational institutions may need dedicated servers or powerful workstations to support multiple concurrent users. In addition, maintaining and updating virtual machine environments can increase administrative complexity and operational costs.

2.2.3.2 Container-Based Training

Container-based training environments utilize containerization technologies to create isolated and lightweight learning environments. Compared to traditional virtual machines, containers generally consume fewer resources and can be deployed more efficiently.

This approach has become increasingly popular in cybersecurity education due to its scalability and reduced infrastructure overhead. Learners can access isolated environments that support practical cybersecurity activities while minimizing system resource consumption.

Despite these advantages, container-based environments still require technical setup, deployment management, and infrastructure maintenance. Furthermore, learners may need to understand additional concepts related to containerization technologies, which can increase complexity for beginners who are primarily focused on developing investigation skills.

2.2.3.3 Traditional E-Learning Platforms

Traditional e-learning platforms and Learning Management Systems (LMS) focus primarily on delivering educational content through lectures, reading materials, quizzes, and assessments. These systems are commonly used in academic environments due to their accessibility and ease of implementation.

While traditional e-learning platforms are effective for knowledge delivery and theoretical learning, they often provide limited opportunities for practical investigation activities and hands-on experience. Learners typically consume information passively rather than actively engaging with realistic cybersecurity scenarios.

As a result, students may develop strong theoretical understanding while lacking practical experience in applying investigative techniques within simulated cybersecurity environments.

2.2.3.4 Simulation-Based Virtual Filesystem Approach

The approach adopted in KINETIC BREACH is a simulation-based training environment built around a virtual terminal and a database-driven virtual filesystem. Instead of relying on real operating systems, virtual machines, or containerized infrastructures, the system simulates investigation activities within a controlled browser-accessible environment.

Users interact with a Linux-like terminal interface and investigate cybersecurity incidents by examining logs, exploring files, and discovering digital evidence stored within a virtual filesystem. The environment provides realistic investigation experiences while eliminating the security risks and infrastructure complexity associated with traditional cybersecurity laboratories.

This approach offers several advantages. First, it significantly reduces hardware and infrastructure requirements because no virtual machines or containers are created for individual users. Second, the browser-based design improves accessibility by allowing learners to access the platform using standard web browsers. Third, the simulated environment maintains complete control over investigation content and progression, enabling the integration of adaptive guidance, gamification, and structured learning activities.

Furthermore, the simulation-based approach aligns with the project's objective of providing a beginner-friendly environment that focuses specifically on log analysis and system investigation skills rather than advanced infrastructure management.

2.2.3.5 Technique Comparison and Selected Approach

Table 2.2 summarizes the comparison between the reviewed approaches.

Table 2.2: Comparison of Training Delivery Environments

Feature	VM	Container	LMS	KINETIC BREACH
Realism	High	High	Low	Medium
Resource Usage	High	Medium	Low	Low
Accessibility	Medium	Medium	High	High
Infrastructure Complexity	High	Medium	Low	Low
Investigation Training	High	High	Low	High
Beginner Friendly	Medium	Medium	High	High

The simulation-based virtual filesystem approach was selected for KINETIC BREACH because it provides an effective balance between educational value, accessibility, implementation feasibility, and learner engagement. While virtual machines and containerized environments offer greater realism, they also introduce additional infrastructure complexity and maintenance requirements. In contrast, the selected approach enables learners to develop practical investigation skills through realistic simulations while maintaining a lightweight, scalable, and accessible educational platform.

Therefore, the simulation-based virtual filesystem approach was determined to be the most suitable technique for achieving the objectives of the proposed system.

2.3 Project Methodology

The development of KINETIC BREACH follows the Agile Software Development Methodology. Agile is an iterative and flexible software development approach that emphasizes continuous improvement, collaboration, adaptability, and incremental development throughout the software development lifecycle. Unlike traditional linear development models, Agile encourages developers to continuously evaluate progress, incorporate feedback, and refine system functionalities through multiple development iterations.

The Agile methodology was selected for this project because KINETIC BREACH consists of multiple interconnected components, including a virtual terminal environment, investigation scenarios, a discovery-based investigation framework, AI-assisted guidance mechanisms, gamification features, and user progression systems. Due to

the complexity and evolving nature of these components, it was difficult to define all system requirements and implementation details at the beginning of the project.

Furthermore, several system components such as the AI-assisted guidance system and investigation framework required continuous testing, evaluation, and refinement throughout development. Therefore, Agile was considered the most suitable methodology as it supports incremental implementation, iterative improvement, and continuous testing while allowing flexibility when introducing new features or modifying existing functionalities.

Figure 2.1 illustrates the Agile Software Development Methodology adopted throughout the development of KINETIC BREACH.

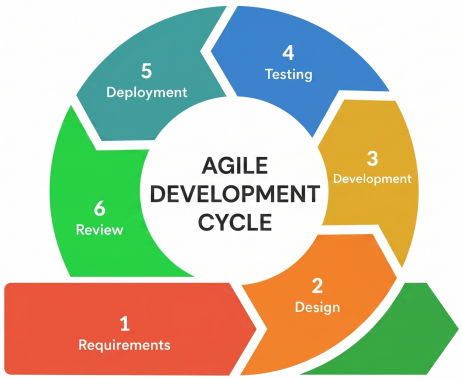


Figure 2.1: Agile Software Development Methodology

The development process was divided into several iterations, where each iteration focused on delivering specific functionalities while continuously improving the system based on testing outcomes and implementation progress.

2.3.1 Requirements Gathering and Planning

The first phase focused on identifying the objectives, scope, functional requirements, and non-functional requirements of the project. During this stage, literature reviews, existing system analysis, and domain studies were conducted to understand the challenges associated with cybersecurity education and practical investigation training.

The findings obtained from the research activities were used to define the core objectives of KINETIC BREACH, determine the target users, identify suitable technologies, and establish the overall direction of the project. Initial system requirements related to investigation scenarios, learner interaction, gamification features, and AI-assisted guidance were also documented during this phase.

2.3.2 Iteration 1: Core Infrastructure

The first development iteration focused on establishing the fundamental infrastructure required for the system. This included the implementation of user authentication, user account management, database design, and core backend services.

During this phase, the PostgreSQL database structure was developed to support user information, scenario data, investigation progress, scoring records, and system configurations. Basic frontend and backend communication mechanisms were also established to provide a stable foundation for subsequent development iterations.

2.3.3 Iteration 2: Terminal and Virtual Environment

The second iteration focused on the development of the virtual investigation environment. A browser-based terminal interface was implemented to simulate Linux command-line interactions within a controlled educational environment.

In addition, a database-driven virtual filesystem was developed to store files, directories, logs, and digital artefacts required for investigation activities. This approach allowed learners to perform realistic investigation tasks without requiring virtual machines, containers, or real operating system environments.

The implementation of the virtual environment represented one of the core technical components of KINETIC BREACH and served as the foundation for all investigation activities within the platform.

2.3.4 Iteration 3: Investigation Engine

The third iteration focused on developing the investigation engine and scenario framework. During this phase, the cybersecurity investigation scenarios were designed and integrated into the system.

The implementation included evidence discovery mechanisms, objective tracking, scenario progression logic, and investigation validation processes. Multiple investigation paths were supported to encourage exploration and independent problem-solving rather than relying solely on predefined solution paths.

The three investigation scenarios implemented in the project are Brute Force Login Investigation, Suspicious Script Execution Investigation, and SSH Forensics Investigation.

2.3.5 Iteration 4: AI and Adaptive Guidance

The fourth iteration introduced the AI-assisted guidance component of the system. This phase focused on providing contextual support to learners who encounter difficulties during investigation activities.

The guidance mechanism analyzes user behaviour and investigation progress to determine whether assistance is required. Based on the learner's current situation, contextual hints are generated to help users continue their investigations without directly revealing solutions.

This iterative implementation process allowed the guidance system to be continuously refined through testing and evaluation to ensure that assistance remained helpful while preserving the educational value of independent investigation.

2.3.6 Iteration 5: Gamification and Refinement

The fifth iteration focused on enhancing learner engagement through gamification features and overall system refinement. Various motivational mechanisms were integrated into the platform, including scoring systems, experience points (XP), achievements, rankings, and performance tracking.

Additional improvements were made to the user interface, scenario flow, investigation experience, and system usability. Feedback obtained from testing activities was used to refine existing functionalities and improve the overall learning experience provided by the platform.

This phase also involved optimizing system performance, improving user interaction workflows, and ensuring consistency across all implemented modules.

2.3.7 Testing and Evaluation

Testing and evaluation activities were conducted continuously throughout all development iterations. Functional testing was performed to verify that individual system components operated according to their intended requirements, while integration testing ensured proper communication between different modules.

In addition, usability evaluation was conducted to assess user interaction, learner experience, investigation workflow effectiveness, and overall system performance. The collected feedback was used to identify potential improvements and validate whether the project objectives had been successfully achieved.

The iterative nature of Agile development enabled testing results to be incor-

porated continuously into subsequent development cycles, resulting in progressive improvements throughout the implementation process.

2.4 Project Requirements

This section describes the software, hardware, and additional resources required for the development, testing, and evaluation of KINETIC BREACH. The identified requirements support the implementation of the web-based gamified cybersecurity training platform, including frontend and backend development, database management, AI integration, testing activities, and project documentation.

The requirements are categorized into software requirements, hardware requirements, and other supporting requirements.

Table 2.3: Software Requirements

Software	**Category**	**Purpose**
Visual Studio Code	IDE	Used for frontend and backend source code development.
Node.js	Runtime Environment	Executes the backend application and server-side logic.
Express.js	Backend Framework	Provides RESTful API and backend service implementation.
React.js	Frontend Framework	Develops the interactive user interface of the platform.
PostgreSQL	Database	Stores users, scenarios, progress, evidence, and system data.
pgAdmin 4	Database Management	Manages and monitors PostgreSQL databases.
Git & GitHub	Version Control	Tracks source code changes and project versions.
Google Chrome	Web Browser	Used for system testing and execution.
Postman	API Testing Tool	Tests and validates backend API endpoints.
OpenAI API	AI Service	Provides contextual hint generation and adaptive guidance.

Table 2.4: Hardware Requirements

Hardware	Purpose
Personal Computer or Laptop	Used for software development, database management, testing, and report preparation.
Internet Connection	Required for accessing online resources, API services, GitHub repositories, and documentation.
Storage Device or Cloud Backup	Used for project backup, source code storage, and document management.

Table 2.5: Other Requirements

Requirement	Purpose
Academic Journals and References	Support literature review and research activities.
Supervisor Consultation Sessions	Provide guidance, feedback, and project monitoring.
Presentation Facilities	Support project presentations and demonstrations.
User Testing Environment	Support usability testing and evaluation activities.
Documentation Tools	Support report writing and project documentation.

2.5 Project Schedule and Milestones

A project schedule was developed to organize and monitor the development activities of KINETIC BREACH throughout the project duration. The schedule was designed to ensure that all project objectives, development tasks, testing activities, and documentation requirements could be completed within the allocated timeframe.

The project was divided into several phases, including requirements gathering, literature review, system design, implementation, testing and evaluation, documentation, and final project preparation. Each phase contains specific milestones that contribute to the successful completion of the project.

Figure 2.2 presents the project schedule and milestones for the development of KINETIC BREACH.

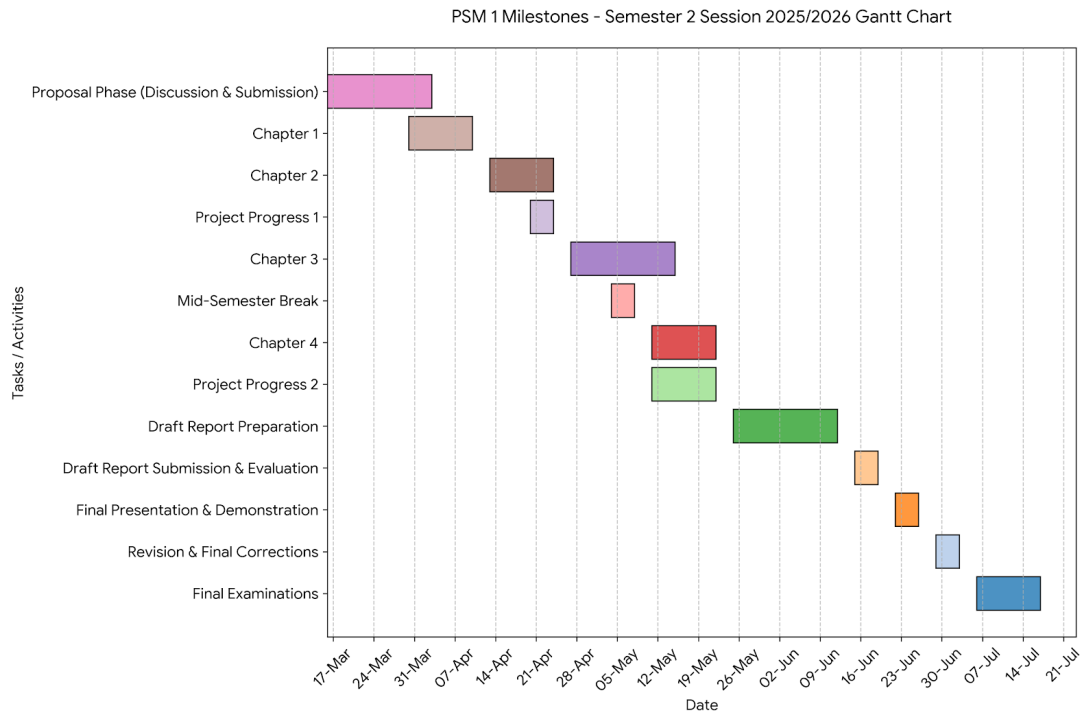


Figure 2.2: Project Schedule and Milestones for KINETIC BREACH

The project schedule was used as a guideline throughout the development process to track progress, manage development activities, and ensure timely completion of project deliverables.

2.6 Conclusion

This chapter presented the facts, findings, methodologies, and technical foundations that support the development of KINETIC BREACH. The discussion examined several key domains relevant to the project, including cybersecurity education and training, gamification in education, interactive and hands-on learning, simulation-based cybersecurity training, artificial intelligence assisted learning support, and digital forensics and log analysis training.

Furthermore, existing cybersecurity training platforms were reviewed and analyzed to identify their strengths, limitations, and educational approaches. The comparison highlighted the differences between existing systems and KINETIC BREACH, particularly in terms of investigation-based learning, AI-assisted guidance, and the use of a lightweight simulation-based environment for practical cybersecurity training.

This chapter also justified the selection of the simulation-based virtual filesystem approach and discussed the Agile Software Development Methodology adopted throughout the project. In addition, the software requirements, hardware requirements,

supporting resources, project schedule, and development milestones necessary for the successful implementation of the system were presented.

The knowledge and findings obtained from this chapter provide the theoretical and technical foundation for the design and implementation of KINETIC BREACH. The next chapter focuses on the system analysis and design, including the overall system architecture, database design, system modules, user interaction flow, and detailed design specifications of the proposed platform.

CHAPTER 3

ANALYSIS

3.1 Introduction

This chapter presents the analysis phase of KINETIC BREACH, a gamified training system designed to support the development of log analysis and system investigation skills among undergraduate students. The purpose of the analysis phase is to examine the existing learning environment, identify the challenges associated with current cybersecurity training approaches, and determine the requirements necessary for the development of the proposed system.

The analysis process focuses on understanding the needs of learners who require practical experience in cybersecurity investigation activities while maintaining accessibility and ease of use. Existing educational approaches, learning limitations, and training challenges are examined to identify the gaps that influence the development of practical investigative skills. Particular attention is given to issues such as limited hands-on investigation opportunities, passive learning approaches, and the complexity of many existing cybersecurity training environments.

In addition, this chapter identifies the data, functional, non-functional, and supporting requirements required to implement the proposed platform. These requirements define the information that must be managed by the system, the functionalities that must be provided to users and administrators, and the operational characteristics necessary to ensure a reliable and effective learning experience.

The findings obtained from this analysis phase provide the foundation for the design and implementation of KINETIC BREACH. The identified requirements and analysis outcomes will be used to guide the system architecture, database design, user interface development, and module implementation discussed in the following chapter.

3.2 Problem Analysis

3.2.1 Current Learning Environment

Current cybersecurity learning environments in many undergraduate Information Technology (IT) and software development programmes primarily emphasize theoretical learning through lectures, reading materials, quizzes, and guided laboratory activities. These educational approaches provide learners with foundational cybersecurity knowledge but often offer limited opportunities to develop practical investigation skills through realistic and interactive learning experiences.

In many cases, students learn cybersecurity concepts by attending lectures, studying educational materials, completing guided laboratory exercises, and taking assessments designed to evaluate theoretical understanding. While these activities contribute to knowledge acquisition, they may not adequately prepare learners for real-world cybersecurity investigation tasks that require evidence discovery, log analysis, analytical thinking, and independent problem-solving.

Furthermore, practical activities within traditional learning environments are frequently highly structured and instructor-driven. Learners are often provided with predefined instructions and expected outcomes, reducing opportunities for exploration and independent investigation. As a result, students may complete cybersecurity-related coursework without gaining sufficient experience in investigating realistic system incidents or analyzing suspicious activities commonly encountered in defensive cybersecurity roles.

Another challenge is that many existing cybersecurity training platforms either focus on passive learning approaches or require learners to interact with complex virtualized environments. Such environments may introduce technical barriers that can discourage beginner learners and increase the learning curve associated with practical cybersecurity training.

Figure 3.1 illustrates a typical cybersecurity learning process commonly found in traditional educational environments and highlights several limitations that may affect learner engagement and practical skill development.

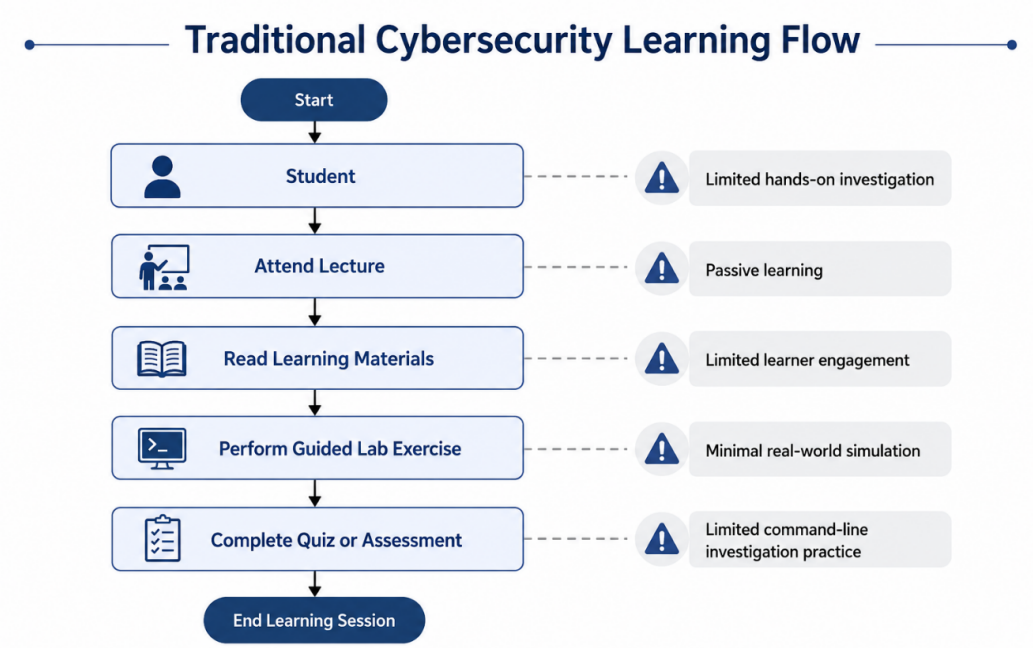


Figure 3.1: Traditional Cybersecurity Learning Flow

3.2.2 Identified Problems

Based on the analysis of current cybersecurity learning environments, several issues have been identified that may limit the development of practical investigation skills among undergraduate students. These issues highlight the need for a more interactive, accessible, and investigation-oriented training platform that can support both learner engagement and practical skill development.

3.2.2.1 PS1: Lack of Practical Investigative Training

Many undergraduate Information Technology (IT) and software development programmes emphasize theoretical instruction and guided laboratory exercises rather than realistic investigative activities. Although students are introduced to cybersecurity concepts through lectures, tutorials, and assessments, they often receive limited exposure to practical investigation tasks that require independent analysis and evidence discovery.

As a result, learners may possess theoretical knowledge of cybersecurity concepts while lacking hands-on experience in analyzing authentication logs, investigating suspicious system behaviour, identifying indicators of compromise, and correlating evidence during cybersecurity investigations.

3.2.2.2 PS2: Gap in Entry-Level Defensive Cybersecurity Skills

Entry-level cybersecurity positions frequently require individuals to perform activities such as log analysis, incident identification, evidence collection, and system investigation. However, many students graduate without sufficient practical experience in these areas due to limited opportunities to engage with realistic investigation scenarios during their studies.

This gap between academic learning and practical cybersecurity requirements may reduce learner confidence and make it more difficult for graduates to apply their knowledge within professional cybersecurity environments. Consequently, additional training is often required before learners can effectively perform investigation-related tasks.

3.2.2.3 PS3: Passive Learning and Low Learner Engagement

Traditional educational approaches commonly rely on passive learning methods such as lectures, reading materials, and quizzes. While these methods are effective for delivering theoretical knowledge, they often provide limited opportunities for active participation, exploration, and problem-solving.

The lack of interactive learning activities may reduce learner engagement and motivation throughout the learning process. Furthermore, passive learning environments provide fewer opportunities for students to develop analytical thinking and decision-making skills that are essential for cybersecurity investigations.

3.2.2.4 PS4: Complexity of Existing Cybersecurity Training Environments

Many practical cybersecurity training platforms depend on virtual machines, cyber ranges, containerized infrastructures, or complex laboratory environments. Although these platforms can provide realistic experiences, they often require significant hardware resources, technical configuration, and infrastructure management.

For beginner learners, the complexity of these environments may create additional learning barriers and distract from the primary objective of developing investigation skills. Educational institutions may also face challenges related to deployment, maintenance, and scalability when implementing such training solutions.

Overall, the identified problems demonstrate the need for a beginner-friendly and investigation-focused training platform that provides practical learning experiences while maintaining accessibility, learner engagement, and manageable infrastructure requirements. These findings serve as the foundation for the requirement analysis

presented in the following section.

3.3 Requirement Analysis

3.3.1 Data Requirement

Data requirements define the information that must be collected, processed, stored, and retrieved by the system to support its intended functionalities. For KINETIC BREACH, several categories of data are required to facilitate user management, investigation activities, learner progression, scenario execution, and AI-assisted guidance.

The system manages both static and dynamic data throughout the learning process. Static data includes user profiles, investigation scenarios, objectives, virtual files, and system configurations. Dynamic data includes investigation progress, command execution history, discovered evidence, learner achievements, and AI interaction records generated during active learning sessions.

The identified data requirements are summarized in Table 3.1.

Table 3.1: Data Requirements of KINETIC BREACH

Data Category	**Description**
User Data	Stores user profile information, authentication credentials, and account details.
Scenario Data	Stores investigation scenarios, objectives, mission briefings, and completion criteria.
Virtual Filesystem Data	Stores virtual files, directories, logs, evidence artefacts, and simulated system content.
Session Data	Stores active investigation sessions, learner progress, and scenario states.
Command History Data	Stores terminal commands executed during investigations for analysis and progress tracking.
Discovery Data	Stores discovered evidence, completed objectives, and investigation milestones.
Progression Data	Stores scores, XP, achievements, rankings, and performance records.
AI Interaction Data	Stores hint requests, generated hints, and adaptive guidance records.

The relationships between the major data entities used within KINETIC BREACH are illustrated in Figure **??**.

The data model provides the foundation for storing investigation information, managing learner activities, tracking progression, and supporting adaptive guidance functionalities throughout the platform.

3.3.2 Functional Requirement

Functional requirements describe the operations and services that must be provided by KINETIC BREACH to support cybersecurity investigation training activities. These requirements define how the system interacts with users, processes investigation activities, manages learner progression, and supports adaptive guidance throughout the learning experience.

The primary functional requirements of the proposed system are described as follows:

FR1: User Registration and Authentication

The system shall allow users to create accounts, authenticate their credentials, and securely access the platform. User sessions shall be managed to ensure controlled access to learning resources and investigation activities.

FR2: Scenario Selection and Mission Access

The system shall allow learners to view and select available investigation scenarios. Each scenario shall provide mission briefings, objectives, and contextual information before the investigation begins.

FR3: Simulated Investigation Environment

The system shall provide a browser-based terminal environment that simulates command-line interactions within a controlled investigation setting. Learners shall be able to explore virtual files, logs, and digital evidence through supported commands.

FR4: Terminal Command Processing

The system shall process commands entered by learners and generate appropriate simulated responses based on the active investigation scenario and virtual filesystem content.

FR5: Evidence Discovery and Objective Tracking

The system shall record discovered evidence, completed objectives, and investigation milestones. Learners shall receive progress updates as they advance through investigation activities.

FR6: Investigation Progress Tracking

The system shall continuously monitor learner activities, scenario completion status, command execution history, and overall investigation progress.

FR7: Adaptive Guidance and Hint Support

The system shall provide contextual guidance and hints when learners experience difficulties during investigation activities. Guidance shall assist learners without directly revealing the correct solution.

FR8: Gamification and Progression System

The system shall support gamification features including scoring mechanisms, experience points (XP), achievements, rankings, and learner progression tracking.

FR9: Evaluation and Performance Assessment

The system shall evaluate learner performance based on investigation activities, evidence discovery, objective completion, and overall scenario outcomes.

FR10: Administrative Management

The system shall provide administrative functionalities that allow administrators to manage user accounts, monitor learner activities, maintain investigation scenarios, and oversee overall platform operation.

Figure 3.2 illustrates the interactions between the system actors and the primary functionalities provided by KINETIC BREACH.

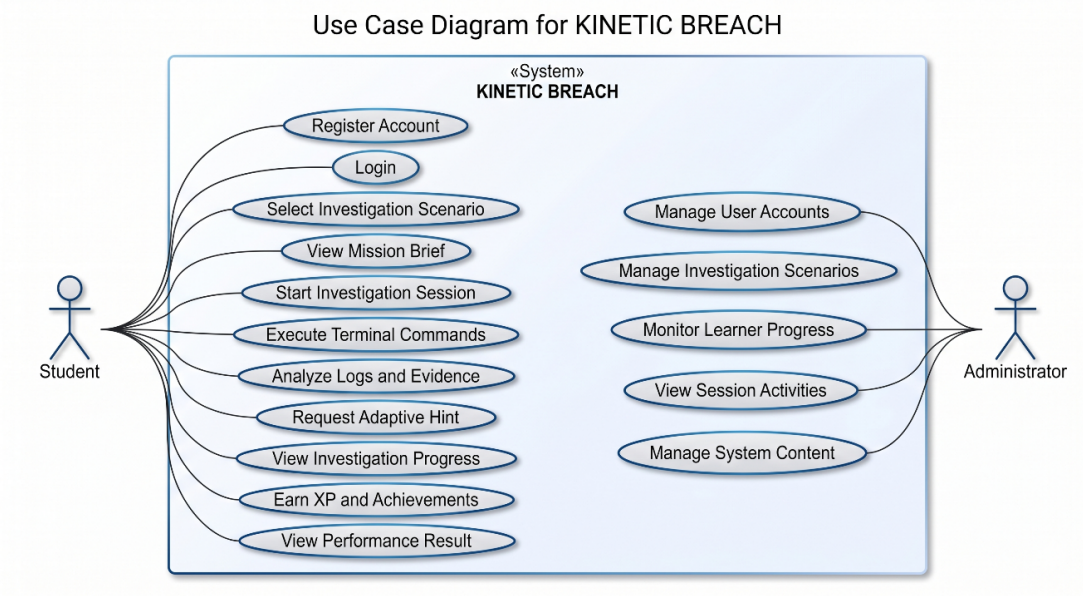


Figure 3.2: Use Case Diagram of KINETIC BREACH

The identified functional requirements define the core operations required to

support practical cybersecurity investigation training within the proposed platform. These requirements serve as the basis for the design and implementation of the system modules discussed in the following chapter.

3.3.3 Non-Functional Requirement

Non-functional requirements define the quality attributes and operational characteristics that KINETIC BREACH must satisfy to provide an effective, reliable, and accessible learning environment. Unlike functional requirements, which describe what the system does, non-functional requirements describe how well the system performs its intended functions.

The identified non-functional requirements focus on usability, performance, reliability, security, accessibility, maintainability, scalability, and compatibility. These requirements ensure that the platform remains suitable for undergraduate learners while supporting practical cybersecurity investigation activities.

Table 3.2 summarizes the non-functional requirements of KINETIC BREACH.

Table 3.2: Non-Functional Requirements of KINETIC BREACH

Requirement	Description
Usability	The system shall provide a beginner-friendly interface that allows learners to navigate investigation activities with minimal training.
Performance	The system shall process terminal commands and investigation activities with minimal response delay to ensure smooth interaction.
Reliability	The system shall maintain stable operation and preserve learner progress, scores, and investigation records without unexpected data loss.
Security	The system shall provide secure authentication, role-based access control, and protection of user data and investigation records.
Accessibility	The system shall be accessible through modern web browsers without requiring additional software installation or advanced configuration.
Maintainability	The system shall support future enhancements, scenario expansion, and feature updates through a modular architecture.
Scalability	The system shall support additional users, investigation scenarios, and educational content while maintaining acceptable performance.
Compatibility	The system shall operate correctly on standard desktop environments and modern web browsers commonly used by students.

The identified requirements provide the operational standards that guide the design, implementation, and evaluation of the proposed system. Meeting these requirements is essential to ensure a stable and effective cybersecurity training environment.

3.4 Other Requirements

In addition to the functional and non-functional requirements, several supporting software, hardware, and development requirements are necessary to ensure the successful implementation of KINETIC BREACH. These requirements were selected

based on the objectives of the project, the nature of the simulation-based investigation environment, and the need to provide an accessible and engaging learning platform for undergraduate students.

3.4.1 Software Requirements

Several software technologies were selected to support the development and operation of KINETIC BREACH. Each technology was chosen based on its suitability for implementing the required system functionalities.

Table 3.3: Software Requirements and Justification

Software	**Justification**
React.js	Selected for developing a responsive and interactive user interface suitable for investigation-based learning activities.
Node.js	Provides an efficient event-driven runtime environment for backend services and API processing.
Express.js	Used to simplify backend development and RESTful API implementation.
PostgreSQL	Chosen for its reliability, scalability, and ability to manage structured investigation and progression data efficiently.
pgAdmin 4	Used for PostgreSQL database administration, monitoring, and maintenance.
Visual Studio Code	Provides a lightweight and feature-rich development environment for frontend and backend development.
Git and GitHub	Used for version control, source code management, and project backup.
xterm.js	Provides a realistic browser-based terminal interface that supports simulated command-line investigations.
OpenAI API	Supports adaptive guidance and contextual hint generation during investigation activities.
Postman	Used to test and validate API endpoints during development and integration testing.

3.4.2 Hardware Requirements

The proposed system requires standard computing hardware for development, testing, deployment, and evaluation activities. Since KINETIC BREACH uses a browser-based simulation environment rather than real virtual machines or cyber ranges, the hardware requirements remain relatively moderate.

Table 3.4: Hardware Requirements and Justification

Hardware	**Justification**
Personal Computer or Laptop	Required for software development, testing, database management, and documentation activities.
Internet Connection	Required for accessing online resources, API services, source code repositories, and cloud-based services.
Storage Device or Cloud Backup	Used for storing project files, source code, databases, and backup copies of development work.

3.4.3 Additional Supporting Requirements

Apart from software and hardware resources, several supporting requirements are necessary throughout the project lifecycle. These resources assist in research, development, testing, documentation, and project evaluation activities.

Table 3.5: Supporting Requirements

Requirement	**Justification**
Academic References	Support literature review, domain research, and project validation activities.
Supervisor Consultation	Provides guidance, feedback, and project monitoring throughout development.
Testing Environment	Required for functional testing, usability evaluation, and system validation.
Documentation Tools	Support report writing, diagram creation, and project documentation activities.
Presentation Facilities	Required for project demonstration, assessment, and final evaluation.

The identified requirements provide the necessary foundation for the successful development, deployment, and maintenance of KINETIC BREACH. The selected

technologies and resources support the implementation of a lightweight, accessible, and investigation-focused cybersecurity training platform.

3.5 Conclusion

This chapter presented the analysis phase of KINETIC BREACH by examining the limitations of current cybersecurity learning environments and identifying the requirements necessary for the development of the proposed platform. The analysis highlighted several challenges faced by undergraduate students, including limited opportunities for practical investigation training, gaps in entry-level defensive cybersecurity skills, passive learning approaches, and the complexity of many existing cybersecurity training environments.

Furthermore, the chapter identified the data requirements, functional requirements, non-functional requirements, and supporting requirements needed to support the implementation of the proposed system. These requirements define the information managed by the platform, the services provided to learners and administrators, and the operational characteristics required to ensure an effective and reliable learning experience.

The findings obtained from this analysis phase establish a clear foundation for the development of KINETIC BREACH. The identified requirements will guide the design and implementation of the system architecture, database structure, investigation environment, adaptive guidance mechanisms, gamification features, and user interface components.

The next chapter focuses on the design phase of KINETIC BREACH. It presents the overall system architecture, database design, module design, user interface design, and other design specifications that translate the identified requirements into a complete system blueprint for implementation.

CHAPTER 4

DESIGN

4.1 Introduction

This chapter presents the design phase of KINETIC BREACH. The purpose of this phase is to transform the requirements identified during the analysis stage into a complete system design that serves as the blueprint for implementation.

The chapter describes the overall structure of the proposed platform, including the high-level system design, system architecture, user interface design, and database design. These design components define how the various modules interact, how data is managed throughout the platform, and how users engage with the simulated cybersecurity investigation environment.

In addition, the chapter explains the navigation flow, input mechanisms, output presentation, and database organization required to support investigation activities, learner progression, gamification features, and adaptive guidance functionalities. The design specifications presented in this chapter provide the foundation for the implementation activities discussed in the subsequent chapter.

4.2 High-Level Design

The high-level design of KINETIC BREACH provides an overview of the major system components and their relationships before examining the detailed implementation architecture. The system adopts a client-server architecture that separates the user interface, application logic, and data management layers into distinct components.

The frontend component is responsible for presenting the user interface and supporting learner interactions throughout the investigation process. Users access the platform through a web browser where they can authenticate, select investigation scenarios, interact with the virtual terminal environment, receive guidance, and monitor

their progression.

The backend component manages business logic, investigation processing, scenario execution, progression tracking, gamification functionalities, and adaptive guidance services. It acts as the central processing layer that coordinates communication between the frontend and the database.

The database component stores all persistent system information, including user accounts, investigation scenarios, virtual filesystem data, learner progression records, discoveries, achievements, and AI interaction logs. This centralized storage enables consistent management of learner activities and investigation data.

Figure 4.X presents the high-level design of KINETIC BREACH.

[Figure Here: High-Level Design Diagram]

The separation of responsibilities between the frontend, backend, and database components improves maintainability, scalability, and system organization while supporting future expansion of the platform.

4.3 System Architecture

The system architecture of KINETIC BREACH follows a layered client-server architecture that organizes the platform into multiple logical layers. This architectural approach promotes modularity, maintainability, and separation of concerns by assigning specific responsibilities to each layer.

The presentation layer consists of the React.js frontend application. This layer manages user interaction, interface rendering, scenario presentation, virtual terminal interaction, progress visualization, and communication with backend services.

The application layer consists of backend services implemented using Node.js and Express.js. This layer processes user requests, manages investigation logic, validates actions, tracks progression, coordinates adaptive guidance mechanisms, and enforces system rules.

The service layer contains specialized modules responsible for handling investigation activities, command processing, evidence discovery, evaluation mechanisms, AI-assisted guidance, and gamification functionalities. These services operate independently while collaborating to support the overall investigation experience.

The data layer consists of the PostgreSQL database, which stores all persistent system information including user data, scenario data, virtual filesystem content, progression records, discoveries, achievements, and system configuration data.

Figure 4.X illustrates the layered architecture adopted in KINETIC BREACH.

[Figure Here: System Architecture Diagram]

The layered architecture improves system maintainability by isolating different responsibilities and reducing dependencies between components. It also facilitates future enhancements by allowing individual modules to be modified without significantly affecting other parts of the system.

4.3.1 Layered Architecture Design

4.3.2 Static View

4.3.3 Dynamic View

4.4 User Interface Design

4.4.1 Navigation Design

4.4.2 Input Design

4.4.3 Output Design

4.5 Database Design

4.5.1 Conceptual and Logical Database Design

4.5.2 Data Dictionary and Normalization

4.6 Detailed Design

4.6.1 Terminal Parser Design

4.6.2 Player State Analyzer Design

4.6.3 Evaluation Service Design

4.6.4 Discovery Service Design

4.6.5 Hint Level Service Design

4.7 Software Design

4.7.1 Frontend Components

4.7.2 Backend Components

4.7.3 Class Specifications

Bibliography